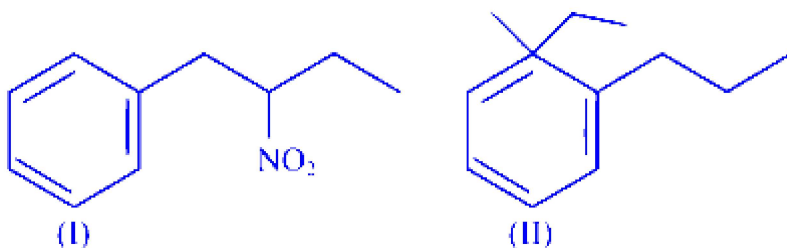


IUPAC Nomenclatures

Question1

IUPAC names of the given compounds (I) and (II) are respectively



AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

5-phenyl-3-nitrobutane, 2-ethyl-2-methyl-1-propylcyclohexane

B.

2-nitro-1-phenylbutane; 2-ethyl-2-methyl-1-propylcyclohexane

C.

2-nitro-1-phenylbutane; 1-ethyl-1-methyl-2-propylcyclohexane

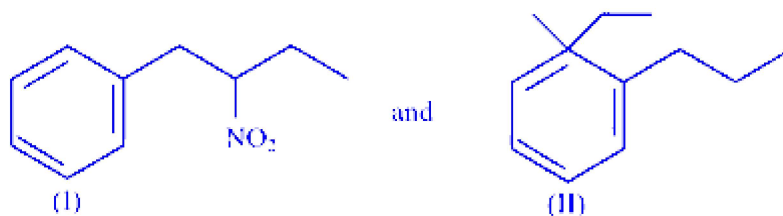
D.

3-nitro-5-phenylbutane; 1-ethyl-1-methyl-2-propylcyclohexane

Answer: C

Solution:

The IUPAC names of



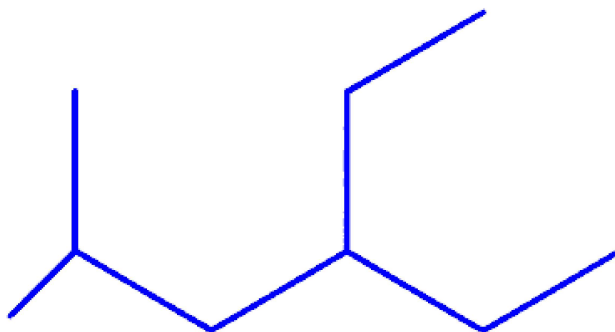
are respectively

2-nitro-1-phenyl-butane

1-ethyl-1-methyl-2-propylcyclohexane

Question2

IUPAC name of the following compound is



AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

2-methyl-4-ethylhexane

B.

4-ethyl-2-methylhexane

C.

5-methyl-3-ethylhexane

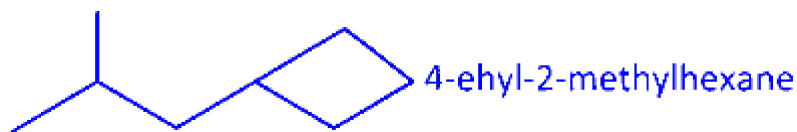
D.

3-ethyl-5-methylhexane

Answer: B

Solution:

The correct IUPAC name of



Question3

IUPAC names of mesityl oxide and oxalic acid are respectively.

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

4 - methylpent - 3 - en - 2 one ; Ethanedioic acid.

B.

4 - methylpent - 3 - en - 2 - one ; Propanedioic acid.

C.

3 - methylpent - 3 - en - 2 - one ; Propanedioic acid.

D.

3 - methylpent - 3 - en - 2 - one ; Ethanedioic acid.

Answer: A

Solution:

The correct IUPAC name of mesityl oxide and oxalic acid are

4-methylpent-3-en-2-one and ethanedioic acid.

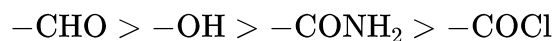
Question4

The correct decreasing order of priority for the functional group of organic compounds in the IUPAC method of nomenclature is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

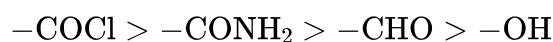
A.



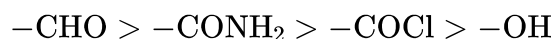
B.



C.



D.

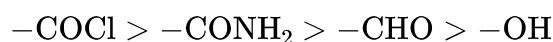


Answer: C

Solution:

The correct order of decreasing priority of functional group is,

Acyl chloride > Amide > Aldehyde > Alcohol



Question5

The IUPAC name of the following compound is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

6-ethyl-9- methyl-4- bromodec-5- en-7- ol

B.

7-bromo-2-methyl-5-ethyldec-5-en-4-ol

C.

7-bromo-5-ethyl-2-methyldec-5-en-4-ol

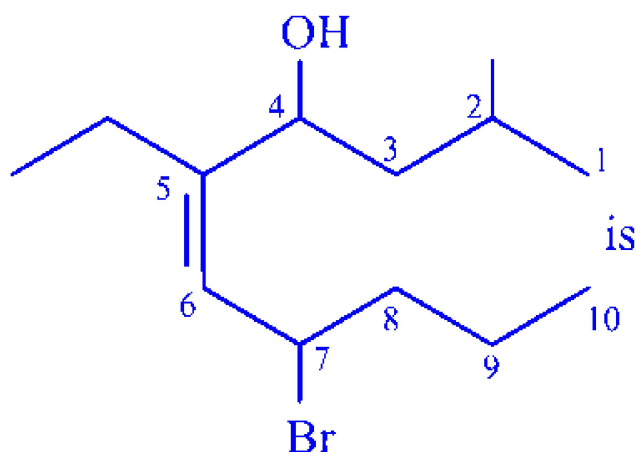
D.

4-bromo-6-ethyl-9-methyldec-5-en-7-ol

Answer: C

Solution:

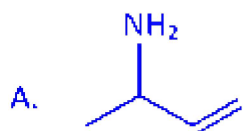
The correct IUPAC name of



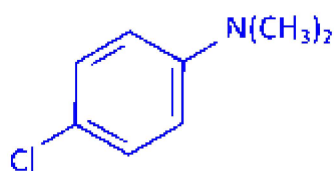
7-bromo-5-ethyl-2 methyl dec-5-en-4-ol

Question6

IUPAC names of the following compounds *A* and *B* are



B.



AP EAPCET 2024 - 21th May Evening Shift

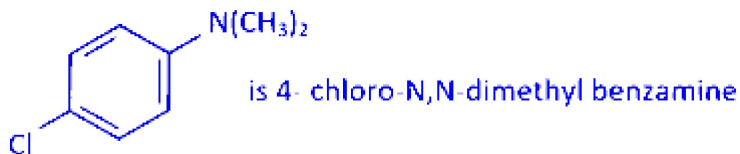
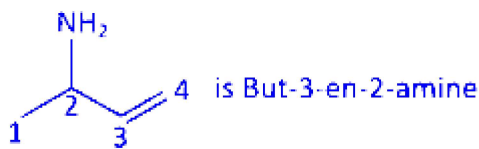
Options:

- A. $A =$ but-3-en-2-amine; $B = 4 - N, N$ -dimethylaminochlorobenzene
- B. $A =$ but-1-en-3-amine; $B = 4 - N, N$ -dimethylaminochlorobenzene
- C. $A =$ but-1-en-3-amine; $B = 4$ -chloro- N, N -dimethylbenzenamine
- D. $A =$ but-3-en-2-amine; $B = 4$ -chloro- N, N -dimethylbenzenamine

Answer: D

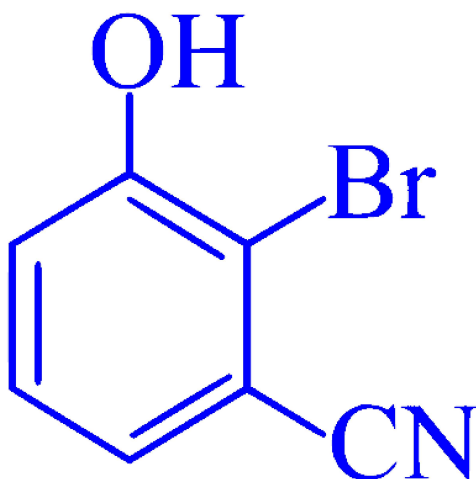
Solution:

The IUPAC name of the given compound i.e.,



Question7

The IUPAC name of the following compound is



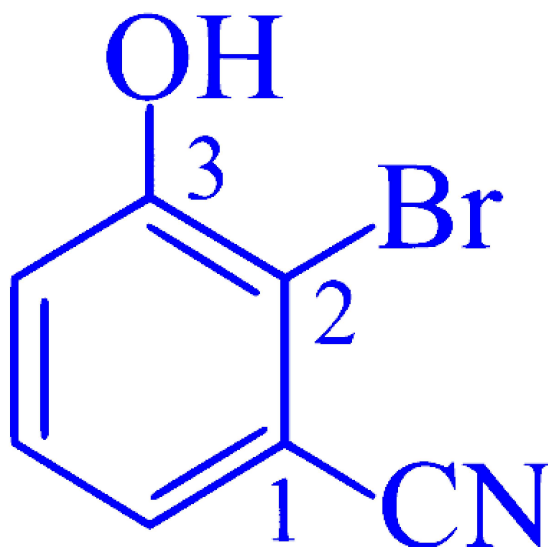
AP EAPCET 2022 - 5th July Morning Shift

Options:

- A. 1-bromo-2-hydroxybenzonitrile
- B. 2-bromo-3-cyanophenol
- C. 2-bromo-3-hydroxy benzonitrile
- D. 3-hydroxy-2-bromobenzonitrile

Answer: C

Solution:



The priority order for the given functionalities is $-\text{CN} > -\text{OH} > -\text{Br}$.

$\therefore$ The numbering will be done from $-\text{CN}$ and in such a way that the groups get the lowest number possible.

Hence, the IUPAC name will be 2-bromo-3-hydroxybenzonitrile.

Question 8

What is the IUPAC name of $\text{CH}_3\text{CH}(\text{CH}_2\text{CH}_3)\text{CHO}$?

AP EAPCET 2021 - 19th August Evening Shift

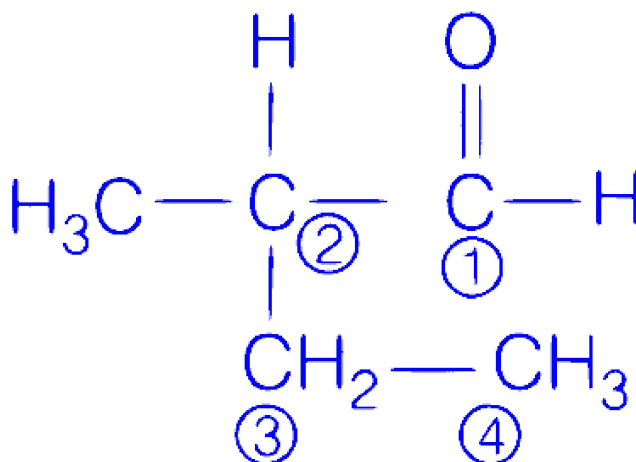
Options:

- A. Butan-2-al
- B. 2-methyl butanal
- C. 3-methylisobutyraldehyde
- D. 2-ethylpropanal

Answer: B

Solution:

—CHO group represent aldehyde group and denoted as 'al'.



Branched chain is —CH₃ which present on 2 carbon position. Principle chain containing 4 carbon with aldehyde group.

IUPAC name-2 methyl butanal.

Question9

The IUPAC name of diacetone alcohol is.

AP EAPCET 2021 - 19th August Evening Shift

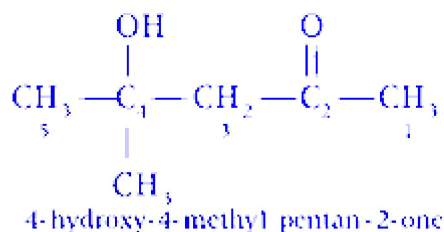
Options:

- A. 4-hydroxy pentan-2, 4-dione
- B. 3-hydroxy pentan-2, 4-dione
- C. 4-hydroxy-4-methyl pentan-2-one
- D. 4-hydroxy-3-methyl pentan-2-one

Answer: C

Solution:

Diacetone alcohol is an organic compound with the formula $\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{C}(\text{OH})(\text{CH}_3)_2$ called DAA. It is colourless liquid and commonly used in synthetic intermediate and as solvent also.



Note $\begin{array}{c} \text{O} \\ || \\ -\text{C}- \end{array}$ is carbonyl ketonic group denoted as 'one'.
